RKNN3 SDK Release Note

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Target Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers

Revision History

Version	Author	Date	Description	Approved By
V0.2.0	HPC	2025-08-16	Initial version	Vincent
V0.3.0b0	HPC	2025-09-12	1. Updated the list of supported models 2. Updated accuracy and performance data	Vincent
v0.4.0b0	HPC	2025-11-03	1. Update the list of supported models 2. Update the overview; model accuracy, and performance data	Vincent
V1.0.0	HPC	2026-01-23	1. Updated the list of supported models 2. Updated accuracy and performance data	Vincent
V1.0.4	HPC	2026-05-15	1. Updated the list of supported models 2. Updated accuracy and performance data 3. Updated supported hardware platforms	Vincent
V1.1.0	HPC	2026-08-19	1. Updated the list of supported models 2. Updated accuracy and performance data	Vincent

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1 Overview

The RKNN3 SDK provides the software stack required to deploy AI models to the RK1820/RK1828 and the RK3572 platform, including a PC development kit (RKNN3 Toolkit), on-device runtime API (RKNN3 Runtime), and model conversion and deployment examples (RKNN3 Model Zoo). This SDK release supports both coprocessor mode (RK1820/1828) and non-coprocessor mode (RK3572). The main differences between the two modes are as follows:

> **[Coprocessor Mode (RK182X)]**

- **Host SoC**: Acts as the system's core, responsible for task scheduling, resource allocation, and overall control.
- **RK1820/RK1828 Coprocessor**: Serves as an AI computation acceleration unit, focusing on high-performance neural network inference tasks.
- **PCIe/USB High-Speed Interface**: Enables low-latency, high-bandwidth data interaction between the host and the coprocessor.

> **[Non-Coprocessor Mode (RK3572)]**

- **RK3572 SoC**: An SoC platform with an integrated high-performance NPU, supporting direct execution of AI inference tasks.
- **Unified Memory Architecture**: The NPU shares memory space with the CPU/GPU.

1.1 Supported Hardware Platforms

- RK1820/RK1828 Coprocessor
- RK3572

1.2 Supported Systems

- Android/Linux/Windows

2 Key Features and Enhancements

- Added RK182X multi-card cascade support
- Added Qwen3.5 model support
- Added Qwen3.5 and Gemma-4 Prefix Caching support
- Added custom CPU operator support
- Added support for different context lengths across sessions
- Added Tie Word Embedding support
- Added RKNN3 Toolkit support for macOS (Beta)
- Optimized host-side memory usage during KVCache import and export
- Optimized `rknn3_mem_sync` API performance
- Optimized GPU memory usage during GRQ quantization
- Updated the usage of external GRQ quantization
- Changed the RKNN3 Toolkit model import method to use ONNX only (TensorFlow/TFLite/Caffe/DarkNet support removed)

3 Supported Models

The currently supported models include but are not limited to the following:

LLM (Large Language Model)

Model Name	Model Source
Qwen2.5-0.5B	https://huggingface.co/Qwen/Qwen2.5-0.5B
Qwen2.5-1.5B	https://huggingface.co/Qwen/Qwen2.5-1.5B
Qwen2.5-3B	https://huggingface.co/Qwen/Qwen2.5-3B-Instruct
Qwen2.5-7B	https://huggingface.co/Qwen/Qwen2.5-7B-Instruct
Qwen3-0.6B	https://huggingface.co/Qwen/Qwen3-0.6B
Qwen3-1.7B	https://huggingface.co/Qwen/Qwen3-1.7B
Qwen3-4B	https://huggingface.co/Qwen/Qwen3-4B
Qwen3-8B	https://huggingface.co/Qwen/Qwen3-8B
Qwen3.5-0.8B	https://huggingface.co/Qwen/Qwen3.5-0.8B
Qwen3.5-2B	https://huggingface.co/Qwen/Qwen3.5-2B
Qwen3.5-4B	https://huggingface.co/Qwen/Qwen3.5-4B
Qwen3.5-9B	https://huggingface.co/Qwen/Qwen3.5-9B
Qwen3.5-27B	https://huggingface.co/Qwen/Qwen3.5-27B
Qwen3.8-27B	https://huggingface.co/Qwen/Qwen3.8-27B
Youtu-LLM-2B	https://huggingface.co/tencent/Youtu-LLM-2B
GLM-Edge-1.5B-Chat	https://modelscope.cn/models/ZhipuAI/glm-edge-1.5b-chat
MiniCPM5-1B	https://huggingface.co/openbmb/MiniCPM5-1B
FunctionGemma-270M	https://huggingface.co/google/functiongemma-270m-it
LFM2.5-1.2B	https://huggingface.co/LiquidAI/LFM2.5-1.2B-Instruct

VLM (Vision-Language Model)

Model Name	Model Source
Qwen2.5-VL-3B	https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct
Qwen2.5-VL-7B	https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct
Qwen3-VL-2B	https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct
Qwen3-VL-4B	https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct
FastVLM-1.5B	https://github.com/apple/ml-fastvlm
InternVL3-2B	https://huggingface.co/OpenGVLab/InternVL3-2B
InternVL3_5-4B	https://huggingface.co/OpenGVLab/InternVL3_5-4B-Instruct
MiniCPM-V-4	https://huggingface.co/openbmb/MiniCPM-V-4
MiMo-VL-7B-RL	https://huggingface.co/XiaomiMiMo/MiMo-VL-7B-RL
LocateAnything-3B	https://huggingface.co/nvidia/LocateAnything-3B
SmolVLM-500M-Instruct	https://huggingface.co/HuggingFaceTB/SmolVLM-500M-Instruct
SmolVLM2-500M-Video-Instruct	https://huggingface.co/HuggingFaceTB/SmolVLM2-500M-Video-Instruct
UI-TARS-2B-SFT	https://huggingface.co/ByteDance-Seed/UI-TARS-2B-SFT
gme-Qwen2-VL-2B-Instruct	https://huggingface.co/Alibaba-NLP/gme-Qwen2-VL-2B-Instruct

Multi-Modal Model

Model Name	Model Source
Qwen2.5-Omni-3B (Thinker)	https://huggingface.co/Qwen/Qwen2.5-Omni-3B
Qwen3-Omni	Commercial closed-source model
Qwen3.5-Omni	Commercial closed-source model
Gemma-4-E2B	https://huggingface.co/google/gemma-4-E2B-it
Gemma-4-E4B	https://huggingface.co/google/gemma-4-E4B-it
Gemma-4-12B	https://huggingface.co/google/gemma-4-12B-it
Gemma-4-31B	https://huggingface.co/google/gemma-4-31B-it

ASR (Speech Recognition)

Model Name	Model Source
Qwen3-ASR-0.6B	https://huggingface.co/Qwen/Qwen3-ASR-0.6B
Whisper	https://huggingface.co/openai/whisper-large-v3
SenseVoiceSmall	https://modelscope.cn/models/iic/SenseVoiceSmall
Zipformer	https://huggingface.co/pfluo/k2fsa-zipformer-chinese-english-mixed

TTS (Text-to-Speech)

Model Name	Model Source
Qwen3-TTS-12Hz-1.7B	https://huggingface.co/Qwen/Qwen3-TTS-12Hz-1.7B-Base
VITS	https://github.com/jaywalnut310/vits

Embedding / Reranker

Model Name	Model Source
Qwen3-Embedding-0.6B	https://huggingface.co/Qwen/Qwen3-Embedding-0.6B
Qwen3-Embedding-4B	https://huggingface.co/Qwen/Qwen3-Embedding-4B
Qwen3-Reranker-0.6B	https://huggingface.co/Qwen/Qwen3-Reranker-0.6B
Qwen3-Reranker-4B	https://huggingface.co/Qwen/Qwen3-Reranker-4B

Translation

Model Name	Model Source
HY-MT1.5-1.8B	https://huggingface.co/tencent/HY-MT1.5-1.8B

OCR

Model Name	Model Source
PaddleOCR VL	https://huggingface.co/PaddlePaddle/PaddleOCR-VL

Computer Vision (CV)

- Image Classification

Model Name	Model Source
MobilenetV2	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/mobilenet/mobilenetv2-12.onnx
Resnet50V2	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/resnet/resnet50-v2-7.onnx

- Object Detection

Model Name	Model Source
YOLOv5s	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/yolov5/yolov5s_rknn3.onnx
YOLOv6s	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/yolov6/yolov6s_rknn3.onnx
YOLOv8s	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/yolov8/yolov8s_rknn3.onnx
YOLO26	https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26s.pt

- Instance Segmentation & Pose Estimation

Model Name	Model Source
YOLO26-Segment	https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26s-seg.pt
YOLO26-Pose	https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26s-pose.pt

- Visual Representation

Model Name	Model Source
Siglip-so400m	https://huggingface.co/google/siglip-so400m-patch14-384
Siglip2-so400m	https://huggingface.co/google/siglip2-so400m-patch14-384
MetaCLIP2	https://huggingface.co/facebook/metaclip-2-worldwide-m16-384
Dinov3	https://huggingface.co/facebook/dinov3-vits16-pretrain-lvd1689m

- Depth Estimation

Model Name	Model Source
Depth-Anything-V2	https://huggingface.co/depth-anything/Depth-Anything-V2-Small
Depth-Anything-V3	https://huggingface.co/depth-anything/DA3-BASE

- Robot Manipulation

Model Name	Model Source
GR00T-N1.6-3B	https://huggingface.co/nvidia/GR00T-N1.6-3B
MiniCPM-RobotManip	https://huggingface.co/openbmb/MiniCPM-RobotManip

Users can download pre-converted RKNN models from the following cloud drive: RKNN3_SDK (https://console.box.lenovo.com/l/H1fig1, Access Code: rknn). The specific path is as follows: `RKNN3_SDK/rknn3_models/v1.1.0`

4 Model Performance

- **LLM Model Performance**

Model Name	Accelerator	Input Tokens	New Tokens	TTFT(ms)	TPOT(ms)	Decode TPS
Qwen2.5-0.5B	RK182X	128	128	21.74	4.46	224.19
Qwen2.5-1.5B	RK182X	128	128	47.25	6.60	151.58
Qwen2.5-3B	RK182X	128	128	83.18	9.62	103.97
Qwen2.5-7B	RK1828	128	128	156.94	14.19	70.47
Qwen3-0.6B	RK182X	128	128	27.74	5.42	184.42
Qwen3-1.7B	RK182X	128	128	52.15	7.10	140.89
Qwen3-4B	RK1828	128	128	105.64	11.22	89.14
Qwen3-8B	RK1828	128	128	175.61	16.15	61.92
Qwen3.5-0.8B	RK182X	128	128	56.54	9.21	108.56
Qwen3.5-2B	RK182X	128	128	79.32	10.94	91.40
Qwen3.5-4B	RK1828	128	128	160.56	19.66	50.86
MiniCPM5-1B	RK182X	128	128	35.69	5.72	174.94
HY-MT1.5-1.8B	RK182X	128	128	58.54	8.04	124.37
GLM-Edge-1.5B-Chat	RK182X	128	128	50.43	6.14	162.81

- **Multi-Card Cascade LLM Model Performance**

Model Name	Accelerator	Input Tokens	New Tokens	TTFT(ms)	TPOT(ms)	Decode TPS
Gemma-4-12B	2x RK1828	5120	128	7576.88	31.85	31.4
Gemma-4-31B	4x RK1828	5120	128	9077.22	70.03	14.28
Qwen3.5-9B	2x RK1828	5120	128	5910.11	30.23	33.08
Qwen3.5-27B	4x RK1828	5120	128	8718.15	76.57	13.06
Qwen3.8-27B	4x RK1828	5120	128	8699.48	76.45	13.08

- **VLM Model Performance**

Model	Accelerator	Vision Resolution	Vision(ms)	TTFT(ms)	Decode TPS
FastVLM-1.5B	RK182X	512 * 512	167.23	47.14	151.41
InternVL3-2B	RK182X	448 * 448	180.67	47.01	149.77
InternVL3_5-4B	RK1828	448 * 448	175.21	106.09	88.94
Qwen2.5-VL-3B	RK182X	784 * 1176	227.43	93.37	51.23
Qwen2.5-VL-3B	RK1828	392 * 392	216.9	83.16	104.76
Qwen2.5-VL-7B	RK1828	784 * 1176	212.29	156.65	70.20
Qwen3-VL-2B	RK182X	384 * 384	105.57	53.73	147.33
Qwen3-VL-4B	RK1828	384 * 384	108.23	106.92	89.95
MiMo-VL-7B-RL	RK1828	784 * 1176	229.99	165.04	65.31
MiniCPM-V-4	RK1828	448 * 448	234.34	94.37	105.59
SmolVLM-500M-Instruct	RK182X	512 * 512	90.60	27.12	157.16
SmolVLM2-500M-Video-Instruct	RK182X	512 * 512	90.27	27.06	157.43
LocateAnything-3B	RK182X	448 * 448	228.70	92.00	102.80

- **Multi-Modal Model Performance**

Model	Accelerator	Vision Resolution	Vision(ms)	Audio(ms)	TTFT(ms)	Decode TPS
Qwen2.5-Omni-3B (Thinker)	RK1828	392 * 392	217.82	97.37	83.47	104.23
Qwen3-Omni	RK1828	384 * 384	149.40	63.60	112.70	86.20
Qwen3.5-Omni	RK1828	384 * 384	91.20	39.00	157.80	62.30
Gemma-4-E2B	RK182X	384 * 384	55.60	102.88	88.16	75.30
Gemma-4-E4B	RK1828	384 * 384	63.20	104.60	149.35	55.33

- **Image Classification Model Performance**

Model Name	Accelerator	Resolution	Single-Core Performance (fps)	Multi-Batch Multi-Core Performance (fps)
MobilenetV2	RK182X	224 * 224	287.78	1361.2
Resnet50V2	RK182X	224 * 224	114.15	847.43

- **Object Detection Model Performance**

Model Name	Accelerator	Resolution	Single-Core Performance (fps)	Multi-Batch Multi-Core Performance (fps)
YOLOv5s	RK182X	640 * 640	35.28	224.26
YOLOv6s	RK182X	640 * 640	30.10	163.24
YOLOv8s	RK182X	640 * 640	34.38	221.79

Note:

1. "RK182X" in the tables indicates that the accelerator chip can be either RK1820 or RK1828.
2. For `Qwen2.5-VL-3B`:

   - When using RK1820 as the accelerator, a two-stage scheme is adopted (the LMHead runs on RK3588).

   - When using RK1828 as the accelerator, the model runs entirely on the coprocessor.
3. The RK1820/RK1828 coprocessor NPU frequency is 1GHz.
4. Tests are based on an RK3588 + RK1820/RK1828 setup connected via PCIe, with the RK3588 set to performance mode.
5. TTFT: Time To First Token.
6. TPOT: Time Per Output Token.
7. TPS: Tokens Per Second.
8. The Vision and LLM timeouts of VLM were tested independently, and the Input Tokens and New Tokens of the LLM part were both set to 128.
9. Multi-card cascade test based on an RK3588 host plus multiple RK1828 coprocessors (using pipeline-parallel cascading), with the RK3588 set to performance mode.

5 Model Accuracy

- **LLM Model Accuracy**

Model Name	Accelerator	Dataset	Orig (float32)	RKNN3 (W4A16 G32)
Qwen2.5-0.5B	RK182X	gsm8k	40.71	36.69
Qwen2.5-1.5B	RK182X	gsm8k	63.08	63.76
Qwen2.5-3B	RK182X	gsm8k	79.91	79.61
Qwen3-0.6B	RK182X	gsm8k	50.72	47.08
Qwen3-1.7B	RK182X	gsm8k	75.36	75.13
Qwen3-4B	RK1828	gsm8k	90.60	90.37
Qwen3.5-4B	RK1828	gsm8k	93.63	92.65

- **VLM Model Accuracy**

Model Name	Dataset	Orig (float32)	RKNN3 (W4A16 G32)
FastVLM-1.5B	MMbench(cn)	58.42	60.57
Qwen2.5-VL-3B	MMbench(cn)	76.8	76.37
Qwen2.5-VL-7B	MMbench(cn)	79.98	81.44
InternVL3-2B	MMbench(cn)	77.23	72.51
InternVL3_5-4B	MMbench(cn)	78.69	77.49
MiMo-VL-7B-RL	MMbench(cn)	74.7	70.36

- **Image Classification Model Accuracy**

Model Name	Dataset	Orig Float32 (TOP1)	Orig Float32 (TOP5)	RKNN3 W8A8 (TOP1)	RKNN3 W8A8 (TOP5)
MobilenetV2	imagenet	0.694	0.888	0.680	0.882
Resnet50V2	imagenet	0.729	0.911	0.720	0.906

- **Object Detection Model Accuracy**

Model Name	Dataset	Orig Float32 AP@0.5:0.95	Orig Float32 AP@0.5	RKNN3 W8A8 AP@0.5:0.95	RKNN3 W8A8 AP@0.5
YOLOv5s	coco2017	0.326	0.481	0.316	0.475
YOLOv6s	coco2017	0.403	0.551	0.386	0.533
YOLOv8s	coco2017	0.39	0.525	0.383	0.517

Note:

1. W4A16 G32 means that weights use 4-bit asymmetric quantization and activations use 16-bit floating point representation. Quantization parameters are assigned for every group of 32 weights along the input channel dimension.
2. W8A8 means that both weights and activations use 8-bit asymmetric quantization.

6 Recommended Server Configuration

The recommended server configurations and conversion time estimates are as follows:

Model Name	Recommended Server Configurations	Estimated Conversion Time
Qwen2.5-0.5B	32-core CPU / 8 GB RAM / 1 TB SSD/HDD	~11 minutes
Qwen2.5-1.5B	32-core CPU / 16 GB RAM / 1 TB SSD/HDD	~26 minutes
Qwen2.5-3B	32-core CPU / 32 GB RAM / 1 TB SSD/HDD	~52 minutes
Qwen2.5-7B	32-core CPU / 64 GB RAM / 1 TB SSD/HDD	~105 minutes

- If you encounter insufficient memory, you can try enabling a Swap partition to expand memory. See reference: https://wiki.debian.org/Swap.

7 References

7.1 Performance Testing Methods

- **General Model Performance Testing**

Reference: **rknn3-runtime/examples/rknn3_model_test_demo/README_CN.md**

- **LLM Model Performance Testing**

Reference: **rknn3-model-zoo/tools/rknn3_llm_test/README.md**

7.2 Accuracy Testing Methods

- **General Model Accuracy Testing**

The rknn3-model-zoo integrates testing methods and code related to image classification and object detection. Users who need to re-verify model accuracy can refer to the instructions shown below.

1. Classification models: **rknn3-model-zoo/examples/mobilenet_v2/README.md**
2. Detection models: **rknn3-model-zoo/examples/yolov8/README.md**
3. **LLM Model Accuracy Testing**

The rknn3-model-zoo also integrates methods and code for LLM model accuracy testing, with support for the CMMLU dataset. For specific testing procedures, refer to **rknn3-model-zoo/tools/rknn3_llm_test/README.md**